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PAPER REVIEWED Sr.No. 4		
PROJECT TITLE: OmniRAG : AN INTELLIGENT MULTIAGENT KNOWLEDGE & PRODUCTIVITY PLATFORM		
GUIDE NAME: Ms. Sweta Sri		
DATE OF SUBMISSION		
1. Title of paper: ReAct: Synergizing Reasoning and Acting in Language Models		
2. Author/s: Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, Yuan Cao (Princeton University, Google Research)		
3. Year of publication: 2023		
4. Journal name/ conference name: 11th International Conference on Learning Representations (ICLR 2023), Kigali, Rwanda		
4.1 Abstract of paper: Large language models are strong at reasoning tasks such as chain-of-thought prompting and at generating action plans for interactive decision-making, but prior work has largely studied these two abilities separately. The authors propose prompting an LLM to produce reasoning traces and task-specific actions in an interleaved sequence, so each supports the other: reasoning helps the model form, track, and revise a plan and recover from unexpected outcomes, while actions let the model query external sources such as a knowledge base, search API, or simulated environment for information the model does not already have. Evaluated on question answering, fact verification, and interactive text-based decision-making benchmarks, this interleaved approach, called ReAct, improves task success and produces more interpretable behavior than prompting with reasoning or actions alone, and it reduces factual errors compared with reasoning that has no access to external grounding.		
4.2 Summary of paper: The paper proposes the ReAct prompting paradigm, in which a language model alternates between generating a short natural-language 'thought' and taking an 'action' — such as searching a knowledge base or interacting with an environment — and then observing the result before continuing. This loop lets the model plan, notice when a retrieved fact does not fit, and adjust its subsequent actions accordingly, rather than committing to a single fixed plan up front. ReAct is tested on		

two categories of tasks: knowledge-intensive reasoning benchmarks (HotpotQA multi-hop question answering and FEVER fact verification), where actions consist of searching and looking up a knowledge source such as Wikipedia, and interactive decision-making environments (ALFWorld and WebShop), where actions directly change the state of a simulated world. Across both categories, ReAct outperforms prompting strategies that use only chain-of-thought reasoning or only action generation. On the knowledge-intensive tasks it produces fewer hallucinated or unsupported claims because each reasoning step can be checked against retrieved evidence, and on the interactive tasks it recovers better from actions that do not go as expected. The authors also show that combining ReAct with chain-of-thought as a fallback, used when retrieval is unhelpful, further improves robustness, and that the resulting thought-action traces are easier for humans to inspect and diagnose than a single opaque chain of reasoning.

5. Key points

5.1 points that are covered in paper :

- The ReAct prompting paradigm: interleaving 'thought' (reasoning) steps with 'action' steps that call an external tool or environment.
- Reduction of hallucination in knowledge-intensive question answering and fact verification by grounding each reasoning step in retrieved observations.
- Evaluation across four benchmarks spanning both knowledge-intensive reasoning (HotpotQA, FEVER) and interactive decision-making (ALFWorld, WebShop).
- Comparison against pure chain-of-thought reasoning and pure action-only prompting baselines.
- Human-interpretable reasoning traces that make it possible to inspect why the model took a given action.

5.2 points that are not covered in paper:

- Coordination between multiple distinct specialized agents — ReAct is a single-agent prompting strategy rather than a multi-agent framework.
- Persistent memory or caching that carries information across separate sessions or follow-up queries.
- A dedicated numerical computation or code-execution capability distinct from general 'action' calls.
- Hierarchical or two-stage retrieval designed for long, structurally rich documents.
- A formal graph-based orchestration layer for managing the thought-action loop at scale.

5.3 Scope for work: • Extend the single-agent thought-action loop into a coordinated multi-agent system, where planning, retrieval, and computation are handled by separate specialized agents, as in OmniRAG's PlanReasonAgent. • Add persistent memory or caching so that reasoning traces and retrieved evidence from earlier queries can be reused for related or follow-up questions. • Combine ReAct-style interleaved reasoning with hierarchical, section-level retrieval to better support question answering over long documents. • Introduce a dedicated code-execution or mathematical-reasoning action type for queries that require calculation rather than lookup. • Study graph-based orchestration of the thought-action loop to make multi-step reasoning more auditable and controllable at scale.

Guide Remark